WEEK 3 · LESSON 16 OF 24 · HANDOUT

How Neural Networks Learn

Study time: **about 50 minutes** (reading, video and activity)

▶ **Watch:** https://www.youtube.com/watch?v=IHZwWFHWa-w — open the lesson online to watch inside the course.

Your goal in this lesson: Explain how a neural network learns using a loss function, gradient descent and backpropagation, and describe the role of the learning rate and epochs.

Learning = reducing mistakes

A new network starts with **random** weights, so its first predictions are nonsense. Training is the process of adjusting millions of weights, little by little, so the predictions become less wrong. Four ideas make this work.

1. The loss function — measuring the mistake

The **loss** is a single number that says how wrong the network is on a set of examples. Small loss = good predictions; big loss = bad ones.

- For regression, a common loss is the **mean squared error** (average of the squared differences).
- For classification, a common loss is **cross-entropy**, which punishes confident wrong answers heavily.

The goal of training is simple to say: **make the loss as small as possible.**

2. Gradient descent — walking downhill

Imagine you are on a hill near Nyungwe in thick fog, and you want to reach the valley. You cannot see far, but you can feel which way the ground slopes under your feet. So you take a step downhill, feel again, take another step, and repeat.

That is **gradient descent**: - The "hill" is the loss. - The **gradient** tells us which direction makes the loss rise fastest — so we step the **opposite** way. - Each step changes every weight a little to reduce the loss.

3. The learning rate — step size

The **learning rate** decides how big each step is.

Learning rate	What happens
Too small	Progress is very slow; training takes forever
Too large	You jump over the valley and bounce around, or even climb up
Just right	Steady progress down to a low loss

Choosing it well is one of the most important settings (**hyperparameters**) in deep learning.

4. Backpropagation — sharing the blame

A network has many layers. When the output is wrong, which weights are to blame? **Backpropagation** answers this. It works **backwards** from the output layer to the first layer, calculating how much each weight contributed to the error.

Then gradient descent adjusts each weight by its share of the blame.

Backpropagation became popular in the 1980s and is still how almost every neural network is trained today.

The training loop

1. Take a small group of examples (a **batch**).
2. **Forward pass:** the network makes predictions.
3. Calculate the **loss**.
4. **Backward pass:** backpropagation computes the gradients.
5. **Update:** gradient descent changes the weights a little.
6. Repeat for all batches. One full pass through the training data is an **epoch**. Training often takes many epochs.

Watch the **validation loss** while training. If training loss keeps falling but validation loss starts rising, the network is overfitting — time for early stopping (Lesson 13).

Key terms

- **Loss** — a number measuring how wrong the predictions are.
- **Gradient** — the direction and steepness of the loss for each weight.
- **Learning rate** — the size of each update step.
- **Batch** — a small group of examples used for one update.
- **Epoch** — one full pass through the training data.
- **Hyperparameter** — a setting chosen by people, not learned (learning rate, number of layers, epochs).

Remember this: Networks learn by measuring their mistakes (loss), finding which way reduces them (gradients via backpropagation), and stepping that way (gradient descent) — many small steps over many epochs.

Name: _____ Date: _____

Activity

Task 1

Explain gradient descent to a friend using a real place you know (a hill, a valley, a staircase) in no more than five sentences.

Task 2

A team's training loss falls quickly, then jumps up and down wildly. What setting would you change first, and how?

Task 3

A dataset has 10,000 examples and the batch size is 100. How many updates happen in one epoch? (Answer: 100.)

Self-check

Think about your answers here, then take the quiz online — it is marked instantly and explains every answer.

1. What does the loss function measure?
   A. How wrong the network's predictions are
   B. The size of the dataset
   C. The number of layers
   D. How fast the computer is
2. In gradient descent, which way do we step?
   A. We do not move
   B. Opposite to the gradient, so the loss goes down
   C. Randomly
   D. In the direction that increases the loss

3. What happens if the learning rate is far too large?

A. The updates overshoot and the loss bounces around or grows

B. Training becomes very slow but steady

C. Nothing changes

D. The model becomes smaller

4. What does backpropagation do?

A. Deploys the model

B. Deletes bad data

C. Works backwards to calculate how much each weight contributed to the error

D. Collects new data

5. What is an epoch?

A. One neuron

B. One full pass through the training data

C. One test set

D. One prediction

6. Training loss falls but validation loss starts rising. What is happening?

A. Perfect learning

B. Underfitting

C. Data leakage is impossible

D. Overfitting — consider early stopping

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